Foundation I: From Source to Manifestation
A Conditional Same-Root Program for Geometry and Modular Flow

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Abstract

Paper I of the Consciousness Trilogy [1] begins from an admissible accessible algebra $(\mathcal{A}_c, \omega)$ and derives the inseparability of geometry and modular flow. The deeper source-side question is how one reaches such an object from the pre-observer substrate developed across Genesis, especially the HAFF and T-DOME papers. The present paper, Foundation I of the Consciousness Series, does not claim a completed unified-source theorem. Instead, it formulates an honest source-side foundation program and pushes its currently closed parts into paper form.

We first state the correct conditional target: if a persistent Nakajima–Zwanzig projection exists in the source datum, and if the foreground-to-algebra transfer can be carried out, then the chain

$$S_{\text{source}} \to \mathcal{K}_* \to \mathfrak{F} \to (\mathcal{A}_c, \omega) \to (\text{Geom}, \text{Mod})$$

is forced. Within that program, we prove the packaging result that budget violation in the persistent record process forces symmetry-broken foreground compression, thereby rewriting the T-DOME I–II chain as a clean Foundation I proposition. We then prove a first restricted PS2 closure in a product-classical, decoherence-dominated pointer regime: pointer coarse-graining yields fragment-local omission losses, semigroup decay yields a uniformly bounded coherence remainder, and a weight gap larger than the resulting remainder forces exact top-k^* pointer selection. In a first finite-dimensional faithful pointer-support regime, the required coherence constants become explicit in Hilbert–Schmidt norm. We then define a pointer-aligned realization of the foreground frame inside the ambient algebra, construct the associated pointer-generated von Neumann algebra, and replace the earlier open-ended transfer candidate by a minimal modular-decoherence stable hull $\Phi^{\text{st}}_{\mathfrak{F}}(\mathfrak{F})$. This stable-hull transfer is well-defined, yields a von Neumann subalgebra, and carries a faithful restricted state.

Finally, under the exact fixed-point form of the HAFF accessibility condition, we prove the inclusion-level comparison

$$\Phi^{\text{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\text{HAFF}},$$

while leaving equality as an open strong-form problem. The upshot is deliberately calibrated: Genesis already supplies substantial antecedent support for Foundation I, but the

completed deductive source-side foundation still depends on two genuinely open interfaces, namely persistent projection selection and the model-independent derivation of PS2 beyond the restricted product-classical pointer regime treated here.

Contents

1 Introduction **4**

2 Source Datum and Object Ledger **5**

3 Target Conditional Theorem and the Source-Selection Problem **6**

4 Conditional Foreground Compression **7**

5 Foreground Realization inside the Ambient Algebra **8**

6 A Restricted PS2 Result in the Decoherence-Dominated Pointer Regime **13**

7 Stable-Hull Transfer and Comparison with the Exact Fixed Sector **27**

8 What Remains Open **30**

9 Conclusion **32**

A External Genesis Inputs (Self-Contained Statements) **33**

B Conditional Input Ledger **34**

References **36**

1 Introduction

Paper I of the Consciousness Trilogy [1] starts at the manifest layer: one assumes an admissible accessible algebra $(\mathcal{A}_c, \omega)$, proves that geometry and modular flow are not independently deformable, and then studies the consequences of that inseparability. That was the correct place for the first trilogy paper to begin, but it leaves a source-foundation question unresolved. What, in the broader Genesis programme, plays the role of the source-side antecedent to $(\mathcal{A}_c, \omega)$?

The present paper, Foundation I of the Consciousness Series, addresses exactly that question. It does not claim that Genesis already contains a completed theorem deriving the trilogy from a single source object. What it does claim is narrower and more precise: the relevant source-side foundation can now be decomposed cleanly, its nearly closed parts can be written as formal propositions, and its actual bottlenecks can be localized without conflating vision and theorem.

The philosophical motivation is correspondingly limited. Paper I showed that geometry and modular flow are not independently deformable once one is already at the manifest pair $(\mathcal{A}_c, \omega)$. Foundation I asks the prior structural question: why should those two readings belong together at all? In the present paper, the same-root idea means only that one seeks a source-side chain from which geometric and modular aspects later arise together. This paper treats that mathematical bridge from source data to accessible algebra; it does not attempt to settle consciousness ontology or the broader philosophy of mind.

The correct three-layer structure is

$$S_{\text{source}} \longrightarrow (\mathcal{A}_c, \omega) \longrightarrow (\text{Geom}, \text{Mod}),$$

with an important internal refinement:

$$S_{\text{source}} \longrightarrow \mathcal{O}_A \longrightarrow \mathcal{O}_B \longrightarrow (\mathcal{A}_c, \omega).$$

Here S_{source} is the pre-observer source datum, $\mathcal{O}_A$ is a persistent projected record process with non-Markovian memory, and $\mathcal{O}_B$ is the symmetry-broken foreground frame produced when full-rank persistence exceeds the available processing budget.

Four claims structure the paper.

1. We state the honest *conditional* Foundation I target. The two genuine open points are persistent projection selection and the model-independent foreground-alignment step PS2 that makes the transfer construction canonical; the existence of an exact fixed-point pointer sector is treated only as an exact-regime PS1 idealization.

2. We isolate a finite-coordinate coding regime in which a binding rank budget and an irreducible symmetry action yield a symmetry-breaking foreground choice. This is a conditional model result, not an import of a universal T-DOME necessity theorem.

3. We prove a first restricted PS2 closure package: in a product-classical, decoherence-dominated pointer regime, pointer coarse-graining yields fragment-local omission losses, semigroup decay yields a uniformly bounded coherence remainder, and a weight gap larger than the resulting remainder forces exact top-k^* pointer selection.

4. We give the paper's main new operator-algebraic construction: a stable-hull realization of the foreground sector. Concretely, we construct a minimal modular-decoherence stable hull, prove that it is a von Neumann subalgebra with faithful restricted state, and prove the first comparison theorem $\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\text{fix}}$ under an explicit fixed-set algebra-closure condition.

This is enough to fix the current calibration. Genesis provides substantial antecedent support for Foundation I. It does not yet provide a finished deductive source-side foundation. No equality between the foreground-generated exact fixed sector, probe-accessible effects, and the environmental record algebra is claimed here.

The paper is organized as follows. Section 2 defines the source datum and the stage outputs. Section 3 states the target conditional theorem and isolates the source-selection problem. Section 4 proves the conditional finite-coordinate compression proposition. Section 5 defines pointer alignment, foreground realization, and the PS2 target. Section 6 proves a first restricted PS2 closure package in the product-classical, decoherence-dominated pointer regime, and records a bookkeeping lemma fixing the diagonal/off-sector split needed for the later model-independent PS2 programme. Section 7 proves the first inclusion-level comparison with the typed exact fixed sector and states a partial Lemma C. Section 8 records the remaining open problems. Appendix A summarizes the external Genesis inputs used in the paper in self-contained mathematical form, and Appendix B collects the named conditional interfaces that remain open.

2 Source Datum and Object Ledger

Definition 2.1 (Source datum)**.** A *source datum* is a quadruple

$$S_{\text{source}} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

where:

1. $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a Type III_1 von Neumann algebra in standard form, with Ω cyclic and separating for $\mathcal{A}_U$;

2. $\{\mathcal{E}_t\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$.

Remark 2.2 (Heisenberg-picture continuity convention). The semigroup $\{\mathcal{E}_t\}$ acts on observables, so the continuity assumption in Definition 2 is pointwise σ-weak continuity on $\mathcal{A}_U$, not norm continuity on the ambient algebra. This is the natural continuity regime for the Heisenberg-picture QDS language used throughout the paper.

The ambient standard-form side is the HAFF input [3]; the projected open-system side is expressed in Nakajima–Zwanzig language [8, 11, 7].

Definition 2.3 (Stage outputs)**.** Given a source datum S_{source}, we use the following ledger.

1. **Stage A output**
$$\mathcal{O}_A := (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s)),$$
where $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$ is a Nakajima–Zwanzig projection, $\omega_{\mathcal{P},t} := \mathcal{P}_*(\omega_t)$ is the projected normal state on the predual, and $\mathcal{K}_*(t,s)$ is the dual memory kernel.

2. **Stage B output**
$$\mathcal{O}_B := (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s), \mathfrak{F}, \pi_{\text{ego}}),$$
where $\mathfrak{F}$ is a k^*-dimensional foreground frame and π_{ego} is the ego prior carried forward from the broken phase of T-DOME II and into the self-calibration analysis of T-DOME III [5, 6].

3. **Stage C output**

$$\mathcal{O}_C := (\mathcal{A}_c, \omega), \qquad \omega = \langle \Omega, \cdot\, \Omega \rangle|_{\mathcal{A}_c}.$$

Remark 2.4 (Stage D). Once an admissible $(\mathcal{A}_c, \omega)$ is available, the Paper I chain from accessible algebra to jointly constrained geometry and modular flow is already in place [1]. The present note is therefore concerned only with Stages A–C.

3 Target Conditional Theorem and the Source-Selection Problem

The honest Foundation I target is conditional and should be stated that way from the start.

Target conditional theorem. Let

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t \geq 0})$$

be a source datum. Assume:

1. there exists a persistent Nakajima–Zwanzig projection $\mathcal{P}$ whose projected predual dynamics carries a non-Markovian kernel $\mathcal{K}_*(t, s)$;
2. a specified finite-coordinate coding model has a binding rank budget and satisfies the optimization and group-action hypotheses of Proposition 4.1, producing a compressed foreground frame $\mathfrak{F}$;
3. there exists a foreground embedding

$$\iota_{\mathfrak{F}} : V_{\text{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\text{sa}}$$

   from which the transfer construction can be carried out and compared, at the inclusion level, with the separately typed exact fixed sector;
4. the resulting $(\mathcal{A}_c, \omega)$ satisfies the admissibility hypotheses needed by Paper I.

Then the chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega) \to (\text{Geom}, \text{Mod})$$

is defined conditional on assumptions (1)–(4). No arrow in this display is asserted to follow from the bare source datum alone.

Remark 3.1 (What is actually open). The conditional theorem has several independent burdens. They include existence and selection of a persistent projection $\mathcal{P}$, the model-specific coding and frame-selection package in Proposition 4.1, and the transfer from the foreground frame to the operator-algebraic stable hull. Paper I shell generation is a further interface input rather than a Foundation I consequence. These burdens must not be compressed into a single claimed necessity chain.

Lemma 3.2 (Conditional source-to-kernel interface)**.** *Let S_{source} be a source datum. Suppose there exists a Nakajima–Zwanzig projection $\mathcal{P}$ such that the projected predual dynamics admits a non-Markovian kernel $\mathcal{K}_*(t, s)$ and yields a persistent record process. Then*

$$\mathcal{O}_A = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t, s))$$

satisfies the input conditions of Stage B.

Proof. This is the conditional form of the source-side interface. The assumed projection already provides the Stage A output. Within the T-DOME I modeling framework [4], the persistence assumption is analyzed in a non-Markovian Nakajima–Zwanzig regime; the present paper imports that framing rather than claiming it as a general open-quantum-systems theorem. The point is not to solve the selection problem constructively, but to separate that problem from the downstream thermodynamic and algebraic consequences. □

Remark 3.3 (Constructive version remains open). Lemma 3.2 is intentionally conditional. The missing constructive question is why a given source datum selects a persistent projection at all. Nothing in the present note should be read as claiming that this observer-emergence problem has been solved.

4 Conditional Foreground Compression

We now isolate a finite-coordinate model in which the foreground choice is a valid consequence of explicit coding assumptions. The revised T-DOME results do not establish that memory is necessary for persistence, that every processing budget is saturated, or that every compression problem must break a symmetry. The proposition below therefore states the extra assumptions directly.

Proposition 4.1 (Conditional finite-coordinate foreground compression)**.** *Let*

$$\mathcal{O}_A = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s))$$

be a persistent record process arising from a source datum S_{source}, carrying a specified real coordinate space V of dimension D. Let $h_\mu > 0$ be the stipulated processing rate per retained coordinate. Let admissible encoders be orthogonal projections Q on V, with processing cost $h_\mu \operatorname{rank} Q$, and assume:

1. *the full-rank encoder is infeasible,*

$$h_\mu D > \dot{\mathcal{C}}_{\text{budget}};$$

2. *the survival-distortion functional attains its minimum on the feasible projections and is strictly improved whenever a feasible projection can be enlarged by one retained coordinate;*

3. *a group G acts irreducibly and orthogonally on V, the survival-distortion functional is G-invariant, and $0 < \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$.*

Then every minimizing encoder has rank

$$k^* = \left\lfloor \frac{\dot{\mathcal{C}}_{\text{budget}}}{h_\mu} \right\rfloor < D.$$

Its range defines a k^*-component foreground frame $\mathfrak{F}$. No such proper nonzero projection is G-invariant, so choosing a minimizer selects a frame and breaks the stipulated G-symmetry.

Proof. Feasibility gives $\operatorname{rank} Q \leq \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor$. By strict improvement under every feasible rank-one enlargement, no minimizer can have smaller rank. Hence every minimizer has rank

$$k^* = \Big\lfloor \frac{\dot{\mathcal{C}}_{\text{budget}}}{h_\mu} \Big\rfloor.$$

The range of a minimizing projection supplies the foreground subspace. If that projection commuted with the irreducible G-representation, its range and kernel would be invariant subspaces. Irreducibility would then force the projection to be either 0 or 1, contrary to $0 < k^* < D$. Thus each minimizing choice breaks the stipulated symmetry. This proof uses only the stated coding model; T-DOME I–II motivate the variables but supply no universal necessity premise here [4, 5]. □

Remark 4.2 (Status of Proposition 4.1). The proposition is an FD/I model result. Its rank charge, strict improvement, and irreducible group action are explicit inputs. In particular, it does not say that persistence requires memory, that a general optimum saturates a budget, or that SSB is unavoidable outside this encoder class.

5 Foreground Realization inside the Ambient Algebra

Lemma C requires a genuinely new object: a way to realize the foreground frame inside the ambient algebra.

Definition 5.1 (Exact and finite-window pointer alignment)**.** A self-adjoint observable $O \in \mathcal{A}_U^{\text{sa}}$ is called *pointer-aligned* with respect to the decoherence semigroup $\{\mathcal{E}_t\}_{t\geq 0}$ if it is exactly fixed:

$$\mathcal{E}_t(O) = O \qquad \forall\, t \geq 0.$$

For an explicitly approximate finite-window statement, define

$$d_\tau(O) := \sup_{0\leq t\leq \tau} \|\mathcal{E}_t(O) - O\|.$$

We call O (ε, τ)-pointer-aligned when $d_\tau(O) \leq \varepsilon$.

Remark 5.2 (Why the former asymptotic definition was not weaker). For a bounded semigroup, the former condition $\lim_{t\to\infty} \|\mathcal{E}_t(O) - O\| = 0$ already implies exact fixation. Indeed, for fixed $s \geq 0$, boundedness and the semigroup law give

$$\mathcal{E}_s(O) = \mathcal{E}_s\Big(\lim_{t\to\infty} \mathcal{E}_t(O)\Big) = \lim_{t\to\infty} \mathcal{E}_{s+t}(O) = O.$$

The finite-window quantity d_τ is used only when an honestly approximate notion is intended. Exact fixation remains an idealization of approximate einselection [9, 10].

Let

$$V_{\text{fg}}(\mathfrak{F}) = \operatorname{span}\{f_1, \ldots, f_{k^*}\}$$

be an orthonormal foreground basis selected by the T-DOME II rate-distortion optimum. Guided by the pointer basis selected in the broken phase [5], choose self-adjoint ambient observables

$$O_1, \ldots, O_{k^*} \in \mathcal{A}_U^{\text{sa}}$$

and define

$$\iota_{\mathfrak{F}}(f_i) := O_i, \qquad \iota_{\mathfrak{F}}\Big(\sum_{i=1}^{k^*} a_i f_i\Big) := \sum_{i=1}^{k^*} a_i O_i.$$

Write

$$\mathfrak{S}_{\mathfrak{F}} := \mathrm{span}\{1, O_1, \ldots, O_{k^*}\} \subset \mathcal{A}_U$$

for the resulting operator system and

$$\mathcal{A}_{\mathfrak{F}} := W^*(\mathfrak{S}_{\mathfrak{F}})$$

for its von Neumann envelope.

Proposition 5.3 (Pointer realization of the foreground sector)**.** *Let $\mathfrak{F}$ be a T-DOME foreground frame with*

$$V_{\mathrm{fg}}(\mathfrak{F}) = \mathrm{span}\{f_1, \ldots, f_{k^*}\}.$$

Assume there exists a family of self-adjoint ambient observables

$$O_1, \ldots, O_{k^*} \in \mathcal{A}_U^{\mathrm{sa}}$$

such that:

1. *$O_1, \ldots, O_{k^*}$ are linearly independent;*
2. *each O_i is pointer-aligned under $\{\mathcal{E}_t\}$;*
3. *the ordering of their survival relevance matches the ordering selected by the T-DOME foreground optimization.*

Then the linear extension $f_i \mapsto O_i$ defines an injective map

$$\iota_{\mathfrak{F}} : V_{\mathrm{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\mathrm{sa}}.$$

Moreover,

$$\mathfrak{S}_{\mathfrak{F}} = \mathrm{span}\{1, O_1, \ldots, O_{k^*}\}$$

is a finite-dimensional operator system, and

$$\mathcal{A}_{\mathfrak{F}} = W^*(\mathfrak{S}_{\mathfrak{F}})$$

is a von Neumann subalgebra of $\mathcal{A}_U$.

Proof. Linear independence of $O_1, \ldots, O_{k^*}$ makes the linear extension of $f_i \mapsto O_i$ injective. Since the O_i are self-adjoint, $\mathfrak{S}_{\mathfrak{F}}$ is self-adjoint and contains the identity, hence is an operator system. By definition, $W^*(\mathfrak{S}_{\mathfrak{F}})$ is the von Neumann algebra generated by that operator system inside $\mathcal{A}_U$. Therefore $\mathcal{A}_{\mathfrak{F}}$ is a von Neumann subalgebra of $\mathcal{A}_U$. □

The next refinement is to replace the bare existence assumption on the ambient observables O_i by a more structured compatibility package that points directly back to T-DOME I and II.

The PS1–PS3 package. The structured foreground-realization input naturally separates into three pieces:

(PS1) in the exact dephasing-dominated regime, the decoherence semigroup admits a commuting exact fixed-point pointer sector inside $\mathcal{A}_U$;

(PS2) the T-DOME sufficient-statistic coordinates agree with the coordinates induced by that pointer sector;

(PS3) the T-DOME rate-distortion optimum selects an ordered subset of pointer channels by survival relevance.

In this decomposition, PS1 is an exact-regime idealization of the decoherence-side input (not a consequence of generic decoherence or quantum Darwinism alone, but of the noiseless-sector limit in which pointer projections are exactly fixed rather than approximately einselected), whereas PS2 is the genuine model-independent bridge bottleneck and PS3 is downstream from it.

Proposition 5.4 (Conditional pointer-sector realization)**.** *Assume:*

(PS1) in the exact dephasing-dominated regime, the decoherence semigroup admits a commuting family of nonzero exact fixed-point projections $\{P_1, \dots, P_D\} \subset \mathrm{Fix}(\mathcal{E})$ *generating a classical pointer sector*

$$\mathcal{A}_{\mathrm{ptr}} := W^*(P_1, \dots, P_D) \subseteq \mathcal{A}_U;$$

(PS2) the sufficient-statistic record coordinates used in T-DOME II are the coordinates induced by this pointer sector through the T-DOME I coarse-graining map to the classical sector;

(PS3) the T-DOME II rate-distortion optimum selects an ordered subset $I^* = \{i_1, \dots, i_{k^*}\} \subseteq \{1, \dots, D\}$ *by survival relevance.*

Let

$$V_{\mathrm{fg}}(\mathfrak{F}) = \mathrm{span}\{f_1, \dots, f_{k^*}\}$$

be ordered compatibly with I^**, and define*

$$\iota_{\mathfrak{F}}(f_j) := P_{i_j}.$$

Then:

1. $\iota_{\mathfrak{F}} : V_{\mathrm{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\mathrm{sa}}$ *is injective;*
2. *the image lies in the exact decoherence fixed-point algebra;*
3. *the induced pointer-generated algebra satisfies*

$$\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\mathrm{ptr}}.$$

Proof. By assumption (1), each P_i is self-adjoint and satisfies $\mathcal{E}_t(P_i) = P_i$ for all $t \geq 0$. Distinct nonzero orthogonal projections are linearly independent, so the assignment $f_j \mapsto P_{i_j}$ extends to an injective linear map $\iota_{\mathfrak{F}}$. Exact fixedness gives the second claim. Since the P_{i_j} lie in the commuting pointer sector $\mathcal{A}_{\mathrm{ptr}}$, the von Neumann algebra generated by $1, P_{i_1}, \dots, P_{i_{k^*}}$ is contained in $\mathcal{A}_{\mathrm{ptr}}$. □

Remark 5.5 (Status of PS1). PS1 is the least problematic part of the package. In decoherence theory and quantum Darwinism, robust classical records are carried by a commuting pointer sector selected by the environment [9, 10]. In the exact fixed-point regime used here, this is precisely the statement that the decoherence semigroup admits a commuting family of exact fixed-point projections generating a classical pointer algebra. HAFF Paper D realizes this exactly in the dephasing-dominated finite-dimensional model, where the diagonal fixed-point algebra is the pointer basis algebra [3]. The real model-independent unresolved content is therefore PS2, with PS3 following once PS2 is established.

The PS2 alignment problem. Proposition 5.4 sharpens the existence assumption on $\iota_{\mathfrak{F}}$, but it does not yet derive the compatibility package from the broken-phase output itself in full generality. Section 6 closes a first restricted version in a product-classical pointer regime. The real unresolved content is therefore the model-independent form of the middle condition: the claim that the T-DOME sufficient-statistic coordinates and the pointer-sector coordinates are the same optimization problem written in two languages. That is the full PS2 problem.

Target lemma (PS2 alignment; open). Let

$$\mathcal{A}_{\text{ptr}} = W^*(P_1, \ldots, P_D)$$

be the pointer sector from Proposition 5.4, and define the pointer-coordinate process

$$p_i(t) := \omega_t(P_i), \qquad i = 1, \ldots, D.$$

For an ordered subset

$$I = \{i_1, \ldots, i_k\} \subseteq \{1, \ldots, D\},$$

write $\mathcal{F}_I$ for the pointer-adapted frame retaining those channels. A natural target theorem is the following.

Assume:

1. the T-DOME sufficient statistics $\{c_i(t)\}_{i=1}^D$ agree, after reindexing, with the pointer-coordinate process $\{p_i(t)\}_{i=1}^D$;
2. there exist nonnegative weights $w_1, \ldots, w_D$ and a constant D_0 such that for every pointer-adapted frame $\mathcal{F}_I$,
$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i;$$
3. admissible frames satisfy $|I| = k^* = \lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$.

Then every survival-distortion minimizer retains exactly the k^* channels of largest weight, up to ties. In particular, the T-DOME foreground directions coincide with the top-ranked pointer directions, and the ordered subset I^* of Proposition 5.4 is selected canonically.

Proof strategy. This target should be attacked by an exchange argument. If an admissible subset I omits some channel a and retains some channel b with $w_a > w_b$, then replacing b by a changes the distortion by

$$D(\mathcal{F}_{I\setminus\{b\}\cup\{a\}}) - D(\mathcal{F}_I) = -w_a + w_b < 0,$$

contradicting optimality. Once the distortion has been reduced to such a channelwise omission law, alignment is immediate. The real analytic burden is therefore to derive the weights w_i from the T-DOME survival functional in a way compatible with the pointer-sector coarse-graining.

First realistic route: approximate PS2b in the decoherence-dominated regime. The exact separability law above is likely too strong as a first target in the fully correlated setting. A more plausible theorem program is to prove approximate separability once decoherence has suppressed the cross-channel terms strongly enough that the pointer sector is approximately classical.

Introduce a decoherence-depth parameter τ_{dec}, and let $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \to 0$ measure the size of the residual cross-channel distortion after coarse-graining beyond that scale. The natural asymptotic target is:

$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i + R_I, \qquad |R_I| \le \varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Under this bound, the same exchange argument gives

$$D(\mathcal{F}_{I\setminus\{b\}\cup\{a\}}) - D(\mathcal{F}_I) \le -(w_a - w_b) + 2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Therefore any exchange across a weight gap larger than $2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$ strictly improves the objective. In particular, if the threshold gap

$$\Delta_{k^*} := w_{(k^*)} - w_{(k^*+1)}$$

exceeds $2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$, then the top-$k^*$ pointer subset is forced, up to ties above the gap.

This is likely the right first Bridge 1 theorem target. It matches the physics of decoherence, which suppresses off-diagonal couplings rather than eliminating them exactly in full generality. Section 6 closes this target in a first restricted product-classical regime. The real analytic task is thus to prove that the T-DOME survival distortion admits such a uniform remainder estimate in the decoherence-dominated regime.

Analytic source of $\varepsilon_{\mathrm{dec}}$. The remainder estimate cannot come from abstract operator algebra alone. It has to be extracted from the concrete T-DOME I survival identity

$$\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}} - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle.$$

In the pointer basis, the natural next step is to decompose the record process and the induced state into diagonal and off-diagonal parts,

$$X(t) = X^{\mathrm{diag}}(t) + X^{\mathrm{off}}(t), \qquad \rho(t) = \rho_{\mathrm{diag}}(t) + \rho_{\mathrm{off}}(t),$$

where the diagonal part is generated by the commuting pointer projections and the off-diagonal part carries the residual coherences.

The target analytic lemma is then the following: in the decoherence-dominated regime, the off-diagonal contribution to the survival identity is uniformly bounded by a function $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \to 0$, so that for every pointer-adapted frame $\mathcal{F}_I$,

$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i + R_I, \qquad |R_I| \le \varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

The likely proof route is:

1. use the decoherence semigroup to obtain exponential or semigroup-gap suppression of the off-pointer sector;
2. propagate that suppression through the three terms in the survival identity;
3. identify the diagonal leading term as an additive pointer-channel omission loss.

This identifies the analytic content that Section 6 closes in a first restricted regime.

6 A Restricted PS2 Result in the Decoherence-Dominated Pointer Regime

The full PS2 alignment problem remains open. Still, a genuine theorem can already be proved in a restricted regime. The point of the present section is not to claim a model-independent PS2 theorem, but to show that in a product-classical, decoherence-dominated pointer regime, the structural reduction to fragment-local omission losses and the estimate reduction to a uniformly small coherence remainder can both be closed explicitly, after which approximate separability and top-k^* selection follow by a completely explicit argument.

Proposition 6.1 (Restricted approximate PS2b in the product-classical pointer regime)**.** *Fix $k \in \{1, \dots, D-1\}$, and let $\mathcal{F}_I$ range over all pointer-adapted frames with $|I| = k$. Assume the T-DOME survival distortion uses the squared loss*

$$\ell(x) = x^2.$$

Suppose that for each admissible I there exist random variables

$$L_i = L_i(\xi), \qquad i = 1, \dots, D,$$

and a coherence remainder $C_I = C_I(\xi)$ such that

$$\mathcal{S}_{\text{full}}(\xi) - \mathcal{S}_{\mathcal{F}_I}(\xi) = \sum_{i \notin I} L_i(\xi) + C_I(\xi)$$

for almost every environmental realization ξ. Assume further:

1. ***Pairwise diagonal orthogonality:***

$$\mathbb{E}_\xi[L_i L_j] = 0 \qquad \textit{for } i \neq j;$$

2. ***Uniform coherence bound:***

$$\|C_I\|_{L^\infty(\xi)} \leq \eta_{\text{dec}}(\tau_{\text{dec}}) \qquad \textit{for every admissible } I;$$

3. ***Finite diagonal size:***

$$\Lambda_k := \sup_{|I|=k} \sum_{i \notin I} \|L_i\|_{L^2(\xi)} < \infty.$$

Define

$$w_i := \|L_i\|^2_{L^2(\xi)}$$

and

$$\varepsilon_{\text{dec}}(\tau_{\text{dec}}) := 2\Lambda_k\, \eta_{\text{dec}}(\tau_{\text{dec}}) + \eta_{\text{dec}}(\tau_{\text{dec}})^2.$$

Then for every admissible I,

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

Consequently, if

$$\Delta_k := w_{(k)} - w_{(k+1)} > 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}),$$

then every minimizer of $D(\mathcal{F}_I)$ retains exactly the top-k pointer channels, up to ties above the gap.

Proof. By definition of the distortion and the squared loss,

$$D(\mathcal{F}_I) = \mathbb{E}_\xi \left[\left(\sum_{i \notin I} L_i + C_I \right)^2 \right].$$

Expanding the square gives

$$D(\mathcal{F}_I) = \sum_{i \notin I} \mathbb{E}_\xi[L_i^2] + 2 \sum_{\substack{i,j \notin I \\ i<j}} \mathbb{E}_\xi[L_i L_j] + 2 \sum_{i \notin I} \mathbb{E}_\xi[L_i C_I] + \mathbb{E}_\xi[C_I^2].$$

The pairwise orthogonality assumption cancels the middle sum, so

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I,$$

where

$$R_I := 2 \sum_{i \notin I} \mathbb{E}_\xi[L_i C_I] + \mathbb{E}_\xi[C_I^2].$$

Because the environmental expectation is taken against a probability measure, $\|C_I\|_{L^2(\xi)} \leq \|C_I\|_{L^\infty(\xi)}$. Hence

$$|\mathbb{E}_\xi[L_i C_I]| \leq \|L_i\|_{L^2(\xi)} \|C_I\|_{L^2(\xi)} \leq \|L_i\|_{L^2(\xi)} \eta_{\text{dec}}(\tau_{\text{dec}}),$$

and therefore

$$|R_I| \leq 2\eta_{\text{dec}}(\tau_{\text{dec}}) \sum_{i \notin I} \|L_i\|_{L^2(\xi)} + \eta_{\text{dec}}(\tau_{\text{dec}})^2 \leq \varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

This proves the approximate separability law.

Now let I be admissible, and suppose $a \notin I$, $b \in I$ with $w_a > w_b$. Writing

$$I' := I \setminus \{b\} \cup \{a\},$$

the additive part changes by

$$\sum_{i \notin I'} w_i - \sum_{i \notin I} w_i = -w_a + w_b.$$

The remainder terms satisfy

$$R_{I'} - R_I \leq |R_{I'}| + |R_I| \leq 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}),$$

so

$$D(\mathcal{F}_{I'}) - D(\mathcal{F}_I) \leq -(w_a - w_b) + 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

Therefore any exchange across a weight gap strictly larger than $2\varepsilon_{\text{dec}}(\tau_{\text{dec}})$ lowers the distortion. If $\Delta_k > 2\varepsilon_{\text{dec}}(\tau_{\text{dec}})$, no minimizer can omit a channel above the threshold while retaining one below it. Hence every minimizer is exactly the top-k pointer subset, up to ties above the gap. □

Lemma 6.2 (Product-classical coarse-graining yields fragment-local omission losses)**.** *Assume that after applying the pointer coarse-graining map Φ, the diagonal classical sector is described by D local channel records*

$$\xi = (\xi_1, \ldots, \xi_D), \qquad \xi_i = (z_i, e_i),$$

with full diagonal law

$$P^{\text{full}}(\xi) = \prod_{i=1}^{D} P_i(z_i, e_i).$$

Let $\mathcal{F}_I$ be an admissible pointer-adapted frame. Assume its diagonal projected law is obtained by retaining the full local channel law on I and replacing each omitted channel by a fixed reference law:

$$P^I(\xi) = \prod_{i \in I} P_i(z_i, e_i) \prod_{i \notin I} Q_i(z_i, e_i).$$

Assume further:

1. *the diagonal survival identity is represented pointwise by*

$$\beta\, \mathcal{S}^{\text{diag}}(\xi) = -\imath(\xi) - d(\xi) - \beta\, \delta h(\xi),$$

   where $\imath$ is the pointwise mutual-information density, d is the pointwise reference-divergence density, and δh is the pointwise control contribution;

2. *the reference divergence is taken with respect to a product reference law*

$$\tau(e) = \prod_{i=1}^{D} \tau_i(e_i);$$

3. *the diagonal control contribution is additive:*

$$\delta h^{\text{full}}(\xi) = \sum_{i=1}^{D} \delta h_i^{\text{full}}(\xi_i),$$

$$\delta h^{I}(\xi) = \sum_{i \in I} \delta h_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \delta h_i^{\text{ref}}(\xi_i).$$

For each channel i, define the local pointwise mutual-information density

$$\imath_i^{\text{full}}(\xi_i) := \log \frac{P_i(z_i, e_i)}{P_i^Z(z_i)\, P_i^E(e_i)}, \qquad \imath_i^{\text{ref}}(\xi_i) := \log \frac{Q_i(z_i, e_i)}{Q_i^Z(z_i)\, Q_i^E(e_i)},$$

and the local reference-divergence density

$$d_i^{\text{full}}(e_i) := \log \frac{P_i^E(e_i)}{\tau_i(e_i)}, \qquad d_i^{\text{ref}}(e_i) := \log \frac{Q_i^E(e_i)}{\tau_i(e_i)}.$$

Then the diagonal survival deficit is fragment-local:

$$\beta\big(\mathcal{S}_{\text{full}}^{\text{diag}}(\xi) - \mathcal{S}_{\mathcal{F}_I}^{\text{diag}}(\xi)\big) = \sum_{i \notin I} \lambda_i(\xi_i),$$

where

$$\lambda_i(\xi_i) := -\big(\imath_i^{\text{full}}(\xi_i) - \imath_i^{\text{ref}}(\xi_i)\big) - \big(d_i^{\text{full}}(e_i) - d_i^{\text{ref}}(e_i)\big) - \beta\big(\delta h_i^{\text{full}}(\xi_i) - \delta h_i^{\text{ref}}(\xi_i)\big).$$

Consequently, the centered variables

$$L_i(\xi_i) := \lambda_i(\xi_i) - \mathbb{E}_\xi[\lambda_i(\xi_i)]$$

are fragment-local omission losses in the sense needed by Lemma 6.4.

Proof. Because the full diagonal law factorizes,

$$P^{\text{full}}(\xi) = \prod_{i=1}^{D} P_i(z_i, e_i), \quad P_Z^{\text{full}}(z) = \prod_{i=1}^{D} P_i^Z(z_i), \quad P_E^{\text{full}}(e) = \prod_{i=1}^{D} P_i^E(e_i).$$

Therefore the full pointwise mutual-information density is additive:

$$\imath^{\text{full}}(\xi) := \log \frac{P^{\text{full}}(\xi)}{P_Z^{\text{full}}(z)\, P_E^{\text{full}}(e)} = \sum_{i=1}^{D} \imath_i^{\text{full}}(\xi_i).$$

The same argument for the projected law gives

$$\imath^I(\xi) = \sum_{i \in I} \imath_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \imath_i^{\text{ref}}(\xi_i).$$

Since the reference law τ is also a product law,

$$d^{\text{full}}(e) := \log \frac{P_E^{\text{full}}(e)}{\tau(e)} = \sum_{i=1}^{D} d_i^{\text{full}}(e_i),$$

and similarly

$$d^I(e) = \sum_{i \in I} d_i^{\text{full}}(e_i) + \sum_{i \notin I} d_i^{\text{ref}}(e_i).$$

The control contribution is additive by assumption:

$$\delta h^{\text{full}}(\xi) = \sum_{i=1}^{D} \delta h_i^{\text{full}}(\xi_i),$$

$$\delta h^I(\xi) = \sum_{i \in I} \delta h_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \delta h_i^{\text{ref}}(\xi_i).$$

Substituting these three decompositions into the pointwise diagonal survival identity gives

$$\begin{aligned}
&\beta\big(\mathcal{S}_{\text{full}}^{\text{diag}}(\xi) - \mathcal{S}_{\mathcal{F}_I}^{\text{diag}}(\xi)\big) \\
&= -\big(\imath^{\text{full}}(\xi) - \imath^I(\xi)\big) - \big(d^{\text{full}}(e) - d^I(e)\big) - \beta\big(\delta h^{\text{full}}(\xi) - \delta h^I(\xi)\big) \\
&= \sum_{i \notin I} \Big[-\big(\imath_i^{\text{full}}(\xi_i) - \imath_i^{\text{ref}}(\xi_i)\big) - \big(d_i^{\text{full}}(e_i) - d_i^{\text{ref}}(e_i)\big) - \beta\big(\delta h_i^{\text{full}}(\xi_i) - \delta h_i^{\text{ref}}(\xi_i)\big) \Big].
\end{aligned}$$

This is exactly the stated fragment-local decomposition. Centering each λ_i produces the centered fragment-local omission losses L_i. □

Remark 6.3 (What Lemma 6.2 closes). Lemma 6.2 is the first theorem-level closure of the structure line PS2a. It does not derive the product-classical pointer regime from the broken-phase output; it shows that once the pointer coarse-graining reaches that regime, fragment-local omission losses follow automatically by additivity of information densities, reference-divergence densities, and diagonal control terms.

Lemma 6.4 (Orthogonality from fragment-local centered channel losses)**.** *Assume the environmental realization admits a pointer-channel factorization*

$$\xi = (\xi_1, \dots, \xi_D)$$

with independent coordinates. Suppose the diagonal omission losses of Proposition 6.1 admit a centered fragment-local realization

$$L_i(\xi) = \lambda_i(\xi_i) - \mathbb{E}_\xi[\lambda_i(\xi_i)], \qquad i = 1, \dots, D,$$

for measurable functions $\lambda_i(\xi_i) \in L^2(\xi_i)$. Then

$$\mathbb{E}_\xi[L_i L_j] = 0 \qquad \textit{for } i \neq j.$$

In particular, the pairwise orthogonality hypothesis of Proposition 6.1 is automatic under independent pointer channels together with centered fragment-local omission losses.

Proof. Because L_i depends only on ξ_i and L_j depends only on ξ_j, the independence of ξ_i and ξ_j implies

$$\mathbb{E}_\xi[L_i L_j] = \mathbb{E}_\xi[L_i]\, \mathbb{E}_\xi[L_j].$$

The centered form of L_i and L_j gives

$$\mathbb{E}_\xi[L_i] = 0 = \mathbb{E}_\xi[L_j],$$

hence $\mathbb{E}_\xi[L_i L_j] = 0$. □

Remark 6.5 (Why the orthogonality hypothesis is natural in the pointer regime). Lemma 6.4 explains why the first hypothesis of Proposition 6.1 is not an arbitrary add-on. On the T-DOME side, the computational-ceiling analysis already treats the active channels as independent rate carriers. On the HAFF/Quantum Darwinism side, accessible classical records are carried by independent environmental fragments. Once the diagonal omission loss localizes to fragment-wise centered channel contributions after coarse-graining, pairwise orthogonality becomes a probability identity rather than an extra dynamical postulate.

What remains open is exactly the next step: deriving that fragment-local centered realization, together with the uniform coherence bound, from the full T-DOME survival identity in a model-independent way.

Lemma 6.6 (Semigroup-decay coherence estimate)**.** *Let*

$$\mathbb{E}_{\text{ptr}} : \mathcal{A}_U \to \mathcal{A}_{\text{ptr}}$$

be a normal conditional expectation onto the pointer algebra, and write

$$Q_{\text{off}} := \text{id} - \mathbb{E}_{\text{ptr}}.$$

For each admissible pointer-adapted frame $\mathcal{F}_I$ and environmental realization ξ, let X_ξ denote the relevant state or record datum entering the survival identity, and decompose the coherence remainder as

$$C_I(\xi) = C_I^{(I)}(\xi) + C_I^{(D)}(\xi) + C_I^{(H)}(\xi),$$

where the three terms correspond respectively to the off-diagonal contributions of

$$-\Delta I(S{:}E), \qquad -\Delta D_{\text{KL}}, \qquad -\beta\, \Delta\langle H_{\text{ctrl}}\rangle.$$

Assume:

1. ***Off-sector semigroup decay:*** *there exists a non-increasing function* $g : [0, \infty) \to [0, \infty)$ *with* $g(\tau) \to 0$ *such that*

$$\sup_{t \geq \tau} \|Q_{\text{off}} \mathcal{E}_t(X)\| \leq g(\tau) \, \|Q_{\text{off}} X\|$$

for every relevant datum X*;*

2. ***Uniform off-sector size:***

$$\sup_{\xi} \|Q_{\text{off}} X_\xi\| \leq B_{\text{off}} < \infty;$$

3. ***Termwise continuity bounds:*** *there exist constants* $L_I, L_D, L_H \geq 0$ *such that for every admissible* I *and every* ξ*,*

$$|C_I^{(I)}(\xi)| \leq L_I \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|,$$

$$|C_I^{(D)}(\xi)| \leq L_D \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|,$$

$$|C_I^{(H)}(\xi)| \leq L_H \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|.$$

Then for every admissible I*,*

$$\|C_I\|_{L^\infty(\xi)} \leq \eta_{\text{dec}}(\tau_{\text{dec}}) := (L_I + L_D + L_H) \, B_{\text{off}} \, g(\tau_{\text{dec}}).$$

In particular, the uniform coherence bound required in Proposition 6.1 holds with this choice of η_{dec}*.*

Proof. By the triangle inequality,

$$|C_I(\xi)| \leq |C_I^{(I)}(\xi)| + |C_I^{(D)}(\xi)| + |C_I^{(H)}(\xi)|.$$

Applying the termwise continuity bounds gives

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|.$$

The off-sector semigroup decay assumption then yields

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \, g(\tau_{\text{dec}}) \, \|Q_{\text{off}} X_\xi\|.$$

Using the uniform off-sector bound,

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \, B_{\text{off}} \, g(\tau_{\text{dec}}).$$

Taking the essential supremum over ξ gives the stated L^∞ bound. □

Remark 6.7 (What Lemma 6.6 closes). Lemma 6.6 is the abstract translation step from standard decoherence-semigroup suppression of the off-pointer sector to the exact η_{dec} object used in Proposition 6.1. It does not yet identify the best norm or derive the constants L_I, L_D, L_H from the full T-DOME survival identity; those remain the true analytic content.

In concrete pointer-diagonal control models, one often has $L_H = 0$, so only the mutual-information and relative-entropy terms contribute to the coherence remainder.

Corollary 6.8 (Hilbert–Schmidt continuity package on a uniformly faithful pointer support)**.** *Assume that, after pointer coarse-graining, the relevant joint system–environment states are supported on a finite-dimensional space*

$$\mathcal{H}_S \otimes \mathcal{H}_E$$

with

$$d_S := \dim \mathcal{H}_S, \qquad d_E := \dim \mathcal{H}_E.$$

Assume there exists $m_* > 0$ such that every relevant joint state ρ_{SE}, every relevant marginal state ρ_S, ρ_E, and the fixed reference state τ_E entering the T-DOME ΔD_{KL} term satisfy

$$\lambda_{\min}(\rho_{SE}), \lambda_{\min}(\rho_S), \lambda_{\min}(\rho_E), \lambda_{\min}(\tau_E) \geq m_*.$$

Write $\|\cdot\|_2$ for the Hilbert–Schmidt norm.
Then:

1. *for any two relevant joint states ρ_{SE}, σ_{SE},*

$$\big|I(\rho_{SE}) - I(\sigma_{SE})\big| \leq 3\sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\,\|\rho_{SE} - \sigma_{SE}\|_2;$$

2. *for any two relevant environment states ρ_E, σ_E,*

$$\big|D_{\mathrm{KL}}(\rho_E\|\tau_E) - D_{\mathrm{KL}}(\sigma_E\|\tau_E)\big| \leq \sqrt{d_E}\,\big(2|\log m_*| + 1\big)\,\|\rho_E - \sigma_E\|_2;$$

3. *for any self-adjoint control Hamiltonian $H_{\mathrm{ctrl}} \in \mathcal{B}(\mathcal{H}_S)$,*

$$\big|\mathrm{Tr}[H_{\mathrm{ctrl}}(\rho_S - \sigma_S)]\big| \leq \|H_{\mathrm{ctrl}}\|_2\,\|\rho_S - \sigma_S\|_2.$$

Consequently, in this restricted finite-dimensional faithful regime, one may take Hilbert–Schmidt continuity constants

$$L_I^{\mathrm{HS}} := 3\sqrt{d_S d_E}\,\big(|\log m_*| + 1\big),$$

$$L_D^{\mathrm{HS}} := \sqrt{d_E}\,\big(2|\log m_*| + 1\big),$$

$$L_H^{\mathrm{HS}} := \|H_{\mathrm{ctrl}}\|_2$$

in Lemma 6.6, after translating the relevant off-sector perturbations into the corresponding Hilbert–Schmidt norms.

Proof. Set

$$S(\rho) := -\,\mathrm{Tr}(\rho \log \rho).$$

On the convex set of density operators with spectrum contained in $[m_*, 1]$, the Fréchet derivative is

$$DS_\rho(H) = -\,\mathrm{Tr}\big(H(\log\rho + \mathbf{1})\big).$$

Hence

$$|DS_\rho(H)| \leq \|H\|_2\,\|\log\rho + \mathbf{1}\|_2 \leq \sqrt{d}\,\big(|\log m_*| + 1\big)\|H\|_2$$

for $d = \dim \mathrm{supp}\,\rho$. Applying the mean-value theorem along the line segment between two states gives

$$|S(\rho) - S(\sigma)| \leq \sqrt{d}\,\big(|\log m_*| + 1\big)\|\rho - \sigma\|_2.$$

For the mutual-information term, use

$$I(\rho_{SE}) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}).$$

The partial trace satisfies the Hilbert–Schmidt bound

$$\|\operatorname{Tr}_E A\|_2 \le \sqrt{d_E}\,\|A\|_2, \qquad \|\operatorname{Tr}_S A\|_2 \le \sqrt{d_S}\,\|A\|_2.$$

Therefore

$$|S(\rho_S) - S(\sigma_S)| \le \sqrt{d_S}\,\big(|\log m_*| + 1\big)\|\rho_S - \sigma_S\|_2 \le \sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\|\rho_{SE} - \sigma_{SE}\|_2,$$

and similarly for the environment marginal. For the joint entropy term,

$$|S(\rho_{SE}) - S(\sigma_{SE})| \le \sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\|\rho_{SE} - \sigma_{SE}\|_2.$$

Summing the three contributions proves part (1).

For the relative-entropy term with fixed faithful reference τ_E, write

$$D_{\mathrm{KL}}(\rho_E\|\tau_E) = \operatorname{Tr}(\rho_E \log \rho_E) - \operatorname{Tr}(\rho_E \log \tau_E).$$

Hence

$$\begin{aligned}&\big|D_{\mathrm{KL}}(\rho_E\|\tau_E) - D_{\mathrm{KL}}(\sigma_E\|\tau_E)\big| \\ &\le \big|S(\rho_E) - S(\sigma_E)\big| + \big|\operatorname{Tr}\big((\rho_E - \sigma_E)\log\tau_E\big)\big|.\end{aligned}$$

The entropy term is bounded by

$$\sqrt{d_E}\,\big(|\log m_*| + 1\big)\|\rho_E - \sigma_E\|_2,$$

while

$$\|\log \tau_E\|_2 \le \sqrt{d_E}\,|\log m_*|.$$

Cauchy–Schwarz therefore gives part (2).

Part (3) is immediate from the Hilbert–Schmidt Cauchy–Schwarz inequality. □

Remark 6.9 (Status of Corollary 6.8). Corollary 6.8 does not close the full estimate track in general Type III form. It closes the first finite-dimensional faithful pointer-support regime, where the Hilbert–Schmidt norm is natural and all three continuity constants become explicit. For the restricted Bridge 1 target developed in the present note, that is enough: the estimate line is closed at theorem level. The further question of whether an equally clean package survives beyond that regime, or must be reformulated in a predual norm, belongs to a later strengthening rather than to the current closure claim.

Corollary 6.10 (Restricted PS2 closure in the product-classical faithful pointer regime)**.** *Assume:*

1. *the hypotheses of Lemma 6.2;*
2. *the independent pointer-channel factorization hypothesis of Lemma 6.4;*
3. *the off-sector semigroup decay and uniform off-sector size hypotheses of Lemma 6.6;*
4. *the finite-dimensional uniformly faithful pointer-support hypotheses of Corollary 6.8;*

5. *the centered fragment-local omission losses from Lemma 6.2 satisfy*

$$\Lambda_k := \sup_{|I|=k} \sum_{i \notin I} \|L_i\|_{L^2(\xi)} < \infty.$$

Define

$$w_i := \|L_i\|^2_{L^2(\xi)},$$
$$\eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) := \left(L^{\mathrm{HS}}_I + L^{\mathrm{HS}}_D + L^{\mathrm{HS}}_H\right) B_{\mathrm{off}}\, g(\tau_{\mathrm{dec}}),$$

and

$$\varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) := 2\Lambda_k\, \eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) + \left(\eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}})\right)^2.$$

Then every admissible pointer-adapted frame $\mathcal{F}_I$ satisfies

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I, \qquad |R_I| \le \varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Consequently, if

$$\Delta_k := w_{(k)} - w_{(k+1)} > 2\varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}),$$

then every minimizer of $D(\mathcal{F}_I)$ retains exactly the top-k pointer channels, up to ties above the gap.

Proof. Lemma 6.2 produces centered fragment-local omission losses L_i. Lemma 6.4 upgrades the independent pointer-channel factorization to the pairwise orthogonality required in Proposition 6.1. By Corollary 6.8, the continuity constants appearing in Lemma 6.6 may be chosen as $L^{\mathrm{HS}}_I, L^{\mathrm{HS}}_D, L^{\mathrm{HS}}_H$, so that Lemma 6.6 yields the uniform coherence bound

$$\|C_I\|_{L^\infty(\xi)} \le \eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

The finiteness hypothesis gives the remaining size bound $\Lambda_k < \infty$. Proposition 6.1 therefore applies and gives exactly the stated approximate separability law and top-k selection criterion. □

Corollary 6.11 (Pure-dephasing envelope)**.** *Assume the hypotheses of Proposition 6.1, and suppose in addition that the coherence remainder is controlled by the omitted pointer-channel coherences in the form*

$$|C_I(\xi, t)| \le \sum_{i \notin I} a_i\, |p_i(t)| \qquad \textit{for all } t \ge \tau_{\mathrm{dec}},$$

where $a_i \ge 0$ are constants and each $p_i(t)$ is the exact Lorentz–Drude pure-dephasing decoherence function from T-DOME I [4, 7],

$$\gamma_i < 4\lambda_i, \qquad \Omega_i := \tfrac{1}{2}\sqrt{4\lambda_i\gamma_i - \gamma_i^2} \in \mathbb{R}_{>0},$$

and

$$p_i(t) = e^{-\gamma_i t/2}\left[\cos(\Omega_i t) + \frac{\gamma_i}{2\Omega_i}\sin(\Omega_i t)\right].$$

Define

$$R_i := \sqrt{1 + \left(\frac{\gamma_i}{2\Omega_i}\right)^2}, \qquad A_k := \max_{|I|=k} \sum_{i \notin I} a_i R_i.$$

Then one may take

$$\eta_{\mathrm{dec}}(\tau_{\mathrm{dec}}) = A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2}, \qquad \gamma_* := \min_{1\le i\le D} \gamma_i,$$

and hence

$$\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \le 2\Lambda_k A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2} + A_k^2\, e^{-\gamma_* \tau_{\mathrm{dec}}}.$$

In particular, the distortion remainder decays exponentially with the decoherence depth.

Proof. For each channel,

$$|p_i(t)| \le e^{-\gamma_i t/2}\sqrt{1 + \left(\frac{\gamma_i}{2\Omega_i}\right)^2} = R_i\, e^{-\gamma_i t/2} \le R_i\, e^{-\gamma_* t/2}$$

because $|a\cos x + b\sin x| \le \sqrt{a^2+b^2}$. Therefore

$$|C_I(\xi,t)| \le \sum_{i\notin I} a_i R_i\, e^{-\gamma_* t/2} \le A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2}$$

for all $t \ge \tau_{\mathrm{dec}}$. This identifies a valid choice of $\eta_{\mathrm{dec}}(\tau_{\mathrm{dec}})$. The stated bound on $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$ is then immediate from Proposition 6.1. □

A bookkeeping split for the model-independent PS2 programme. The restricted results above close the PS2 package only after explicit regime assumptions have been imposed. For the later model-independent programme, one further lemma is useful already now: it fixes the exact place where the off-sector estimates enter the survival deficit.

For $X \in \{\mathrm{full}, \mathcal{F}_I\}$, write the survival identity as

$$\beta\, \mathcal{S}_X = -\Delta I_X - \Delta D_X - \beta\, \Delta H_X,$$

where

$$\Delta D_X := \Delta D_{\mathrm{KL},X}, \qquad \Delta H_X := \Delta\langle H_{\mathrm{ctrl}}\rangle_X.$$

Assume each of the three terms admits a pointer-diagonal/off-sector decomposition,

$$\Delta I_X := \Delta I_X^{\mathrm{diag}} + \Delta I_X^{\mathrm{off}},$$

$$\Delta D_X := \Delta D_X^{\mathrm{diag}} + \Delta D_X^{\mathrm{off}},$$

$$\Delta H_X := \Delta H_X^{\mathrm{diag}} + \Delta H_X^{\mathrm{off}}.$$

Define the pointer-diagonal survival functional and off-sector remainder by

$$\beta\, \mathcal{S}_X^{\mathrm{diag}} := -\Delta I_X^{\mathrm{diag}} - \Delta D_X^{\mathrm{diag}} - \beta\, \Delta H_X^{\mathrm{diag}},$$

$$R_X^{\mathrm{off}} := -\Delta I_X^{\mathrm{off}} - \Delta D_X^{\mathrm{off}} - \beta\, \Delta H_X^{\mathrm{off}}.$$

Lemma 6.12 (Diagonal/off-sector split of the survival deficit)**.** *Let $\mathbb{E}_{\mathrm{ptr}} : \mathcal{A}_U \to \mathcal{A}_{\mathrm{ptr}}$ be the conditional expectation onto the pointer algebra selected by the broken phase, and assume the above termwise decompositions are computed with respect to $\mathbb{E}_{\mathrm{ptr}}$. Then for every admissible pointer-adapted frame $\mathcal{F}_I$,*

$$\beta\big(\mathcal{S}_{\mathrm{full}} - \mathcal{S}_{\mathcal{F}_I}\big) = \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}} - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}\big) + R_I^{\mathrm{off}},$$

where

$$R_I^{\mathrm{off}} := R_{\mathrm{full}}^{\mathrm{off}} - R_{\mathcal{F}_I}^{\mathrm{off}}.$$

Equivalently,

$$\begin{aligned} R_I^{\mathrm{off}} = &- \big(\Delta I_{\mathrm{full}}^{\mathrm{off}} - \Delta I_{\mathcal{F}_I}^{\mathrm{off}}\big) \\ &- \big(\Delta D_{\mathrm{full}}^{\mathrm{off}} - \Delta D_{\mathcal{F}_I}^{\mathrm{off}}\big) \\ &- \beta\big(\Delta H_{\mathrm{full}}^{\mathrm{off}} - \Delta H_{\mathcal{F}_I}^{\mathrm{off}}\big). \end{aligned}$$

Proof. Write the survival identity separately for the full process and the frame-adapted process $\mathcal{F}_I$, insert the termwise diagonal/off-sector decompositions into both identities, and subtract. The pointer-diagonal contributions group into

$$\beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}} - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}\big),$$

while the off-sector contributions group into

$$R_{\mathrm{full}}^{\mathrm{off}} - R_{\mathcal{F}_I}^{\mathrm{off}}.$$

This is exactly the stated identity. □

Remark 6.13 (What this does and does not close). Section 6 is a restricted theorem-level closure of PS2, not a general one. What is now closed is the implication

$$\begin{aligned} \text{uniformly small coherence remainder} &\Longrightarrow \text{approximate separability} \\ &\Longrightarrow \text{top-}k \text{ pointer selection.} \end{aligned}$$

Corollary 6.10 packages this as a single restricted selection theorem in the product-classical and finite-dimensional faithful pointer regime, while Corollary 6.11 gives a model-specific exponential envelope. Lemma 6.12 does not enlarge that closure claim; it only fixes the bookkeeping split showing that the already-controlled semigroup-decay terms belong to the off-sector remainder. What remains open is the model-independent derivation of the regime assumptions, and in particular control of the diagonal product defect, from the full T-DOME survival identity and the broken-phase output.

A conditional model-independent foreground-background package. The bookkeeping split also makes it possible to state an honest conditional version of the next model-independent target. For budget-feasible I with $|I| = k^*$, define the diagonal-tail survival deficit by

$$\Delta_I^{\mathrm{diag}}(\xi, t) := \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}(\xi, t)\big),$$

the corresponding tail distortion by

$$D_\tau^{\mathrm{diag}}(I) := \sup_{t \geq \tau} \mathbb{E}_{P_t^{\mathrm{diag}}}\Big[\big(\Delta_I^{\mathrm{diag}}(\xi, t)\big)^2\Big],$$

and for an exchange pair

$$\mathfrak{P} := I^* \times (I^*)^{\mathrm{c}}, \qquad N_{\mathrm{ex}} := |\mathfrak{P}| = k^*(D - k^*),$$

define

$$\begin{aligned} G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi, t) &:= \beta\big(\mathcal{S}_{\mathcal{F}_{I_{a\leftrightarrow b}}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathcal{F}_{I^*}}^{\mathrm{diag}}(\xi, t)\big), \\ \gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi, t) &:= \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathrm{ff}}^{\mathrm{diag}}(u, t) - \mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(v, t)\big), \\ \overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t) &:= \mathbb{E}_{P_t^{\mathrm{diag}}}\big[\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi, t)\big]. \end{aligned}$$

Proposition 6.14 (Conditional diagonal foreground-background coupling suppression)**.** *Let I^* be the broken-phase foreground set. Assume:*

1. ***Approximate tail optimality.*** *There exists $r_{\mathrm{opt}}(\tau) \to 0$ such that*

$$D_\tau^{\mathrm{diag}}(I^*) \le D_\tau^{\mathrm{diag}}(I) + r_{\mathrm{opt}}(\tau) \qquad \textit{for all } |I| = k^*.$$

2. ***Exchange cover and tail persistence.*** *There exist $c_{\mathrm{cov}} > 0$, $M_\gamma < \infty$, $e_{\mathrm{cov}}(\tau) \to 0$, and $r_{\mathrm{pers}}(\tau) \to 0$ such that:*

$$G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t)\,\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t) \ge 0 \qquad \textit{for all } (a,b) \in \mathfrak{P},\ t \ge \tau,$$

$$\frac{1}{N_{\mathrm{ex}}} \sum_{(a,b)\in\mathfrak{P}} \left|G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t)\right| \ge c_{\mathrm{cov}} \left|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t)\right| - e_{\mathrm{cov}}(\tau) \quad \textit{for a.e. } \xi,\ t \ge \tau,$$

$$\sup_{t\ge\tau} \mathbb{E}\big[|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\cdot,t)|\big] \le M_\gamma,$$

and for

$$\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) := \mathbb{E}\big[G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\,\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\cdot,t)\big]$$

one has

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{s,t\ge\tau} \left|\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) - \Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(s)\right| \le r_{\mathrm{pers}}(\tau).$$

3. ***Tail-passive background survival.*** *There exist $M_{\mathrm{ex}}(\tau) < \infty$ and $\eta_{\mathrm{bb}}(\tau) \to 0$ such that*

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{t\ge\tau} \left\|G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\right\|_{L^1} \le M_{\mathrm{ex}}(\tau),$$

$$\sup_{t\ge\tau} \left\|\mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(\cdot,t)\right\|_{L^\infty} \le \eta_{\mathrm{bb}}(\tau).$$

4. ***Exchange efficiency gap.*** *There exist $M_G(\tau) < \infty$, $m_\gamma(\tau) > 0$, and a constant $\delta_Q > 0$ such that*

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{t\ge\tau} \left\|G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\right\|_{L^\infty} \le M_G(\tau),$$

$$\left|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t)\right| \ge m_\gamma(\tau) \quad \textit{whenever } G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t) \ne 0,$$

and the efficiency ratio

$$\kappa_Q(\tau) := \frac{M_G(\tau)}{m_\gamma(\tau)}$$

satisfies

$$\kappa_Q(\tau) \le 2 - \delta_Q \qquad \textit{for all sufficiently large } \tau.$$

Then there exists a function $\varepsilon_\Gamma(\tau) \to 0$ such that

$$\sup_{t\ge\tau} \left|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)\right| \le \varepsilon_\Gamma(\tau)$$

for all sufficiently large τ. One may take

$$\varepsilon_\Gamma(\tau) := \left[\frac{N_{\mathrm{ex}}}{\delta_Q c_{\mathrm{cov}}} \left(r_{\mathrm{opt}}(\tau) + \frac{2e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma + 2r_{\mathrm{pers}}(\tau) + 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau)\right)\right]^{1/2}.$$

Proof. The exact exchange expansion gives, for every exchange pair,

$$D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*) \le Q_{a\leftrightarrow b}^\tau + 2B_{a\leftrightarrow b}^\tau - 2A_{a\leftrightarrow b}^\tau,$$

where

$$Q_{a\leftrightarrow b}^\tau := \sup_{t\ge\tau} \mathbb{E}\Big[\big(G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\big)^2\Big],$$

$$B_{a\leftrightarrow b}^\tau := \sup_{t\ge\tau} \Big|\beta\,\mathbb{E}\Big[G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\,\mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(\cdot,t)\Big]\Big|,$$

$$A_{a\leftrightarrow b}^\tau := \inf_{t\ge\tau} \Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t).$$

If

$$\sup_{t\ge\tau} \big|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)\big| > \varepsilon,$$

then the exchange-cover and tail-persistence assumptions imply the existence of $(a,b) \in \mathfrak{P}$ with

$$A_{a\leftrightarrow b}^\tau \ge \frac{c_{\mathrm{cov}}}{N_{\mathrm{ex}}}\,\varepsilon^2 - \frac{e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma - r_{\mathrm{pers}}(\tau).$$

The tail-passive background assumption gives

$$2B_{a\leftrightarrow b}^\tau \le 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau).$$

The exchange-efficiency gap implies

$$Q_{a\leftrightarrow b}^\tau \le (2-\delta_Q)\sup_{t\ge\tau}\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) \le (2-\delta_Q)\big(A_{a\leftrightarrow b}^\tau + r_{\mathrm{pers}}(\tau)\big).$$

Substituting these estimates into the exchange bound gives

$$D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*) \le -\frac{\delta_Q c_{\mathrm{cov}}}{N_{\mathrm{ex}}}\,\varepsilon^2 + \frac{2e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma + 2r_{\mathrm{pers}}(\tau) + 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau).$$

Approximate tail optimality of I^* yields

$$-r_{\mathrm{opt}}(\tau) \le D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*).$$

Combining the two inequalities gives the stated bound on ε, hence on $\sup_{t\ge\tau}|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)|$. □

Remark 6.15 (Compatibility and physical nonvacuity of Proposition 6.14). An engineered finite classical Markov witness constructed in the audit layer realizes the four tail packages simultaneously. It establishes their abstract mathematical compatibility, so the antecedent is not formally contradictory. It is not a natural source-datum model and is not used as a physical example here. No known physical source/control model currently derives all four packages jointly, and no known example also realizes the Type III modular shell interface needed downstream. Thus Proposition 6.14 remains a conditional result; its natural physical nonvacuity is open.

Corollary 6.16 (Conditional foreground-background mutual-information decay)**.** *Assume the hypotheses of Proposition 6.14, and in addition suppose that the pointwise diagonal foreground-background defect admits the block-local decomposition*

$$\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t) = -\imath_{\mathrm{fb}}(\xi,t) + e_{\mathrm{fb}}^D(\xi,t) + e_{\mathrm{fb}}^H(\xi,t),$$

where

$$\imath_{\mathrm{fb}}(\xi,t) := \log \frac{P_t^{\mathrm{diag}}(u,v)}{P_t^{\mathrm{diag}}(u)P_t^{\mathrm{diag}}(v)}$$

and the averaged locality defects satisfy

$$\sup_{t\geq\tau}\big|\overline{E}^{D}_{\mathrm{fb}}(t)\big| \leq \varepsilon_D(\tau), \qquad \sup_{t\geq\tau}\big|\overline{E}^{H}_{\mathrm{fb}}(t)\big| \leq \varepsilon_H(\tau),$$

with $\varepsilon_D(\tau), \varepsilon_H(\tau) \to 0$. *Then*

$$\sup_{t\geq\tau} I_t(u:v) \leq \varepsilon_\Gamma(\tau) + \varepsilon_D(\tau) + \varepsilon_H(\tau),$$

hence $I_t(u:v) \to 0$ *uniformly on the tail.*

Proof. Averaging the assumed pointwise decomposition gives

$$\overline{\Gamma}^{\mathrm{diag}}_{\mathrm{fb}}(t) = -I_t(u:v) + \overline{E}^{D}_{\mathrm{fb}}(t) + \overline{E}^{H}_{\mathrm{fb}}(t).$$

Taking absolute values and applying Proposition 6.14 gives the stated estimate. □

Corollary 6.17 (Conditional full diagonal product-defect closure)**.** *Fix orderings*

$$B_{\mathrm{fg}} = (i_1,\dots,i_{k^*}), \qquad B_{\mathrm{bg}} = (j_1,\dots,j_{D-k^*})$$

of the foreground and background indices. For an ordered block $B = (\ell_1,\dots,\ell_r)$*, write*

$$X_{B,\leq m} := (\xi_{\ell_1},\dots,\xi_{\ell_m}), \qquad I_{B,m}(t) := I_t\big(X_{B,\leq m} : \xi_{\ell_{m+1}}\big).$$

Assume the hypotheses of Corollary 6.16, and in addition suppose that for each ordered block $B \in \{B_{\mathrm{fg}}, B_{\mathrm{bg}}\}$ *and each* $m = 1,\dots,|B|-1$ *there is a pointwise diagonal prefix defect*

$$\gamma^{\mathrm{diag}}_{B,m}(x_{B,\leq m+1},t) = -\imath_{B,m}(x_{B,\leq m+1},t) + e^{D}_{B,m}(x_{B,\leq m+1},t) + e^{H}_{B,m}(x_{B,\leq m+1},t),$$

where

$$\imath_{B,m}(x_{B,\leq m+1},t) := \log \frac{P^{\leq m+1}_{B,t}(x_{B,\leq m+1})}{P^{\leq m}_{B,t}(x_{B,\leq m})\,P_{\ell_{m+1},t}(x_{\ell_{m+1}})},$$

and the averaged defects satisfy

$$\sup_{t\geq\tau}\big|\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)\big| \leq \varepsilon^{\Gamma}_{B,m}(\tau),$$

$$\sup_{t\geq\tau}\big|\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]\big| \leq \varepsilon^{D}_{B,m}(\tau), \qquad \sup_{t\geq\tau}\big|\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big]\big| \leq \varepsilon^{H}_{B,m}(\tau),$$

with all these functions tending to zero as $\tau\to\infty$. *Define*

$$\varepsilon^{\mathrm{seq}}_{\mathrm{fg}}(\tau) := \sum_{m=1}^{k^*-1}\Big(\varepsilon^{\Gamma}_{\mathrm{fg},m}(\tau) + \varepsilon^{D}_{\mathrm{fg},m}(\tau) + \varepsilon^{H}_{\mathrm{fg},m}(\tau)\Big),$$

$$\varepsilon^{\mathrm{seq}}_{\mathrm{bg}}(\tau) := \sum_{m=1}^{D-k^*-1}\Big(\varepsilon^{\Gamma}_{\mathrm{bg},m}(\tau) + \varepsilon^{D}_{\mathrm{bg},m}(\tau) + \varepsilon^{H}_{\mathrm{bg},m}(\tau)\Big),$$

with the convention that an empty sum is zero. Then

$$\sup_{t\geq\tau} D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}\middle\|\prod_{i=1}^{D}P_{i,t}\right) \leq \varepsilon_\Gamma(\tau) + \varepsilon_D(\tau) + \varepsilon_H(\tau) + \varepsilon^{\mathrm{seq}}_{\mathrm{fg}}(\tau) + \varepsilon^{\mathrm{seq}}_{\mathrm{bg}}(\tau).$$

Proof. The exact block decomposition gives

$$D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}\middle\|\prod_{i=1}^{D}P_{i,t}\right)=I_t(u:v)+D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}(u)\middle\|\prod_{i\in I^*}P_{i,t}\right)+D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}(v)\middle\|\prod_{j\notin I^*}P_{j,t}\right).$$

For any ordered block $B=(\ell_1,\dots,\ell_r)$, the KL chain rule gives

$$D_{\mathrm{KL}}\left(P_{B,t}\middle\|\prod_{q=1}^{r}P_{\ell_q,t}\right)=\sum_{m=1}^{r-1}I_{B,m}(t).$$

Averaging the assumed prefix decomposition yields

$$\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)=-I_{B,m}(t)+\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]+\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big],$$

hence

$$I_{B,m}(t)\le\left|\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)\right|+\left|\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]\right|+\left|\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big]\right|.$$

Summing over m gives the stated bound for each block. Combining those two blockwise bounds with Corollary 6.16 proves the claim. □

7 Stable-Hull Transfer and Comparison with the Exact Fixed Sector

Let $\omega(\cdot)=\langle\Omega,\cdot\,\Omega\rangle$ on $\mathcal{A}_U$, and let $\sigma_s^{\omega,\mathrm{tot}}$ denote the ambient modular flow. Define

$$\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}:=\left\{\begin{array}{l}\mathcal{M}\subseteq\mathcal{A}_U:\mathcal{M}\text{ is a von Neumann subalgebra,}\\ \qquad\mathcal{A}_{\mathfrak{F}}\subseteq\mathcal{M},\\ \qquad\sigma_s^{\omega,\mathrm{tot}}(\mathcal{M})=\mathcal{M}\quad\forall s\in\mathbb{R},\\ \qquad\mathcal{E}_t(\mathcal{M})\subseteq\mathcal{M}\quad\forall t\ge0\end{array}\right\}.$$

This class is nonempty because $\mathcal{A}_U\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$.

Three distinct structural inputs. For a general normal UCP semigroup, the linear fixed set

$$\mathrm{Fix}(\mathcal{E}):=\{A\in\mathcal{A}_U:\mathcal{E}_t(A)=A\ \forall t\ge0\}$$

is an operator system but need not be a von Neumann algebra under the ambient product. Whenever the comparison results below use

$$\mathcal{A}_{\mathrm{fix}}:=\mathrm{Fix}(\mathcal{E})$$

as an algebra, they assume separately:

(FSA) *fixed-set algebra closure:* $\mathcal{A}_{\mathrm{fix}}$ is a von Neumann subalgebra of $\mathcal{A}_U$;

(MOD) *Type III modular invariance:* $\sigma_s^{\omega,\mathrm{tot}}(\mathcal{A}_{\mathrm{fix}})=\mathcal{A}_{\mathrm{fix}}$ for every $s\in\mathbb{R}$;

(EXP) *expectedness when required:* there is a faithful normal ω-preserving conditional expectation $E_{\mathrm{fix}}:\mathcal{A}_U\to\mathcal{A}_{\mathrm{fix}}$.

FSA is a multiplicative-closure condition and is not supplied by UCP fixedness alone. MOD is a separate dynamical condition. In the present faithful standard-form setting, MOD permits the Takesaki expectation, but the mere existence of a faithful invariant state does not by itself establish MOD or EXP [12, 13]. The minimal stable-hull intersection below does not require EXP; an ω-preserving projection onto the exact fixed sector does.

Definition 7.1 (Stable-hull transfer map)**.** The *stable-hull transfer* of the foreground frame $\mathfrak{F}$ is

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) := \bigcap_{\mathcal{M}\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}} \mathcal{M}.$$

Proposition 7.2 (Preliminary inclusion in the exact fixed sector)**.** *Assume FSA. Thus $\mathcal{A}_{\mathrm{fix}} = \mathrm{Fix}(\mathcal{E})$ is a von Neumann subalgebra, without identifying it with probe accessibility or with an environmental record algebra. Let $\mathfrak{F}$ be a T-DOME foreground frame, and let $O_1, \dots, O_{k^*}$ be exact pointer-aligned observables realizing $\iota_{\mathfrak{F}}$ as in Proposition 5.3. Then*

$$\mathcal{A}_{\mathfrak{F}} = W^*(1, O_1, \dots, O_{k^*}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. Exact pointer alignment gives

$$O_i \in \mathrm{Fix}(\mathcal{E}) = \mathcal{A}_{\mathrm{fix}} \qquad \text{for each } i.$$

Since $\mathcal{A}_{\mathrm{fix}}$ is a von Neumann algebra by FSA, it contains the von Neumann algebra generated by $1, O_1, \dots, O_{k^*}$. Hence

$$W^*(1, O_1, \dots, O_{k^*}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

□

Proposition 7.3 (Minimal stable-hull transfer)**.** *Assume the hypotheses of Proposition 5.3. Then*

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) = \bigcap_{\mathcal{M}\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}} \mathcal{M}$$

is a well-defined von Neumann subalgebra of $\mathcal{A}_U$ satisfying:

1. $\mathcal{A}_{\mathfrak{F}} \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$*;*
2. $$\sigma^{\omega,\mathrm{tot}}_s\big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})\big) = \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \qquad \forall s \in \mathbb{R};$$
3. $$\mathcal{E}_t\big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})\big) \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \qquad \forall t \geq 0;$$
4. *it is minimal with these properties: if $\mathcal{N} \subseteq \mathcal{A}_U$ is any von Neumann subalgebra containing $\mathcal{A}_{\mathfrak{F}}$ and stable under both ambient dynamics, then*
$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{N}.$$

Proof. The class $\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$ is nonempty because $\mathcal{A}_U$ belongs to it. An arbitrary intersection of von Neumann subalgebras is again a von Neumann subalgebra, so the intersection is well-defined and is itself a von Neumann subalgebra of $\mathcal{A}_U$. Since every element of $\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$ contains $\mathcal{A}_{\mathfrak{F}}$, so does the intersection.

If $A \in \Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$, then $A \in \mathcal{M}$ for every $\mathcal{M} \in \mathfrak{L}_{\mathfrak{F}}^{\mathrm{st}}$. By definition of the class, $\sigma_s^{\omega,\mathrm{tot}}(A) \in \mathcal{M}$ for every $s \in \mathbb{R}$, and $\mathcal{E}_t(A) \in \mathcal{M}$ for every $t \geq 0$. Therefore both $\sigma_s^{\omega,\mathrm{tot}}(A)$ and $\mathcal{E}_t(A)$ belong to the intersection. This proves $\sigma_s^{\omega,\mathrm{tot}}(\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})) \subseteq \Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$. Applying the same argument with $-s$ gives the reverse inclusion, hence equality.

Finally, if $\mathcal{N}$ is any von Neumann subalgebra with the same containment and stability properties, then $\mathcal{N} \in \mathfrak{L}_{\mathfrak{F}}^{\mathrm{st}}$. Since $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$ is the intersection of all members of that class, it is contained in $\mathcal{N}$. This is exactly the desired minimality. □

Proposition 7.4 (Stable-hull transfer lies in a modular exact fixed sector)**.** *Assume:*

1. *the hypotheses of Proposition 5.3;*
2. *FSA and the exact pointer-invariance condition of Proposition 7.2;*
3. *MOD holds for* $\mathcal{A}_{\mathrm{fix}}$.

Then

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. By Proposition 7.2, $\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\mathrm{fix}}$. FSA makes $\mathcal{A}_{\mathrm{fix}}$ a von Neumann algebra, MOD makes it invariant under the ambient modular flow, and exact fixedness implies forward stability under the decoherence semigroup. Hence $\mathcal{A}_{\mathrm{fix}} \in \mathfrak{L}_{\mathfrak{F}}^{\mathrm{st}}$. Since $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$ is the intersection of all members of that class, it is contained in $\mathcal{A}_{\mathrm{fix}}$. □

Corollary 7.5 (Partial Lemma C)**.** *Assume the hypotheses of Proposition 7.3. Then* $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$ *is a von Neumann subalgebra of* $\mathcal{A}_U$*, and the restricted vector state*

$$\omega_{\mathfrak{F}}^{\mathrm{st}} := \langle \Omega, \cdot\, \Omega \rangle|_{\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})}$$

is faithful. If, in addition, the hypotheses of Proposition 7.4 hold, then

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. The subalgebra statement is exactly Proposition 7.3. Because Ω is separating for the ambient algebra $\mathcal{A}_U$, it is also separating for every von Neumann subalgebra of $\mathcal{A}_U$; hence the restricted vector state is faithful on $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$. The inclusion claim is Proposition 7.4. □

Anchored shell existence around the stable hull. The stable-hull output already anchors a Foundation I shell problem. Write

$$\mathcal{A}_* := \Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}).$$

For $X \in \mathrm{Ball}(\mathcal{A}_U)$ and $q \geq 0$, define the $\mathcal{A}_*$-relative defect

$$\delta_q^{\mathcal{A}_*,\Omega}(X) := \mathrm{dist}_\Omega^\sharp\big(\mathcal{E}_q(X),\, \mathrm{Ball}(\mathcal{A}_*)\big), \qquad \|Y\|_\Omega^\sharp := \big(\|Y\Omega\|^2 + \|Y^*\Omega\|^2\big)^{1/2}.$$

For $\varepsilon \geq 0$ and $\tau > 0$, let

$$\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega} := \left\{ \mathcal{M} \;\middle|\; \begin{array}{l} \mathcal{A}_* \subseteq \mathcal{M} \subseteq \mathcal{A}_U,\ \mathcal{M} \text{ is a unital vN subalgebra,} \\ \delta_q^{\mathcal{A}_*,\Omega}(X) \leq \varepsilon,\ \forall X \in \mathrm{Ball}(\mathcal{M}),\ \forall q \in \mathbb{Q} \cap [0,\tau] \end{array} \right\}.$$

Proposition 7.6 (Anchored relative-shell existence around the stable hull)**.** *Assume the hypotheses of Corollary 7.5. Then for every* $\varepsilon \geq 0$ *and* $\tau > 0$*:*

1. *the class $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$ is nonempty;*

2. *every chain in $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$ has an upper bound;*

3. *therefore maximal elements exist.*

Proof. By Corollary 7.5, $\mathcal{A}_*$ is a von Neumann subalgebra of $\mathcal{A}_U$ and is forward-stable under the decoherence semigroup. Hence $\mathcal{A}_*$ itself belongs to $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$: for $X \in \mathrm{Ball}(\mathcal{A}_*)$, forward stability gives $\mathcal{E}_q(X) \in \mathcal{A}_*$, and contractivity gives $\mathcal{E}_q(X) \in \mathrm{Ball}(\mathcal{A}_*)$, so $\delta_q^{\mathcal{A}_*,\Omega}(X) = 0$. Thus the class is nonempty.

Let $\{\mathcal{M}_i\}_{i\in I}$ be a chain in $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$, and set

$$\mathcal{M} := W^*\Big(\bigcup_{i\in I} \mathcal{M}_i\Big).$$

Then $\mathcal{A}_* \subseteq \mathcal{M} \subseteq \mathcal{A}_U$, and $\mathcal{M}$ is a unital von Neumann subalgebra. By Kaplansky density, every $X \in \mathrm{Ball}(\mathcal{M})$ is a bounded strong* limit of a net from the unit ball of the algebraic union $\bigcup_i \mathcal{M}_i$. Since the chain is totally ordered, each approximant lies in some admissible member of the chain, so its $\mathcal{A}_*$-relative defect is bounded by ε for all $q \in \mathbb{Q} \cap [0,\tau]$. Passing to the limit in the $\Omega^\sharp$-seminorm yields the same bound for X, hence $\mathcal{M} \in \mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$. Therefore every chain has an upper bound. Zorn's lemma now gives a maximal element. □

Remark 7.7 (Status of anchored shells). Proposition 7.6 does not yet construct the full shell family required on the Paper I side. It closes only the anchored existence problem around the theorem-level Foundation I output $\mathcal{A}_*$. What remains open is the upgrade to a canonical, nontrivial, and calibrated shell family with the correct modular transport law.

Remark 7.8 (Why equality is not the right first target). In general, $\mathcal{A}_{\mathrm{fix}}$ may be much larger than the minimal modular-decoherence stable hull of finitely many foreground pointer observables. The honest first target is therefore the inclusion

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}},$$

not the equality

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \stackrel{?}{=} \mathcal{A}_{\mathrm{fix}}.$$

The equality problem should remain open until one proves that the foreground realization together with the stable-hull transfer generates the full exact fixed sector. A further typed interface would still be needed to relate either object to probe-accessible effects or to the environmental record algebra.

Remark 7.9 (Relation to Paper III). One should not identify the foreground complexity k^* directly with the statistical complexity parameter n of Paper III [2]. The safer statement is only that $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$ provides a natural algebraic candidate for a later bridge to the finite-observer sector studied there.

8 What Remains Open

The bottlenecks are now sharper than they were before this note.

1. **Persistent projection selection.** Lemma 3.2 is only conditional. The constructive question of which projection emerges from the source datum remains open.

2. **Foreground realization and model-independent PS2.** The stable-hull route closes the transfer-map well-definedness problem once $\mathcal{A}_{\mathfrak{F}}$ exists. Within the restricted product-classical and finite-dimensional faithful pointer regime, the PS2 package is now theorem-backed: Lemma 6.2 closes the structural reduction to fragment-local omission losses, Lemma 6.4 closes orthogonality, and Corollary 6.10 packages the restricted selection theorem. Lemma 6.12 now fixes the bookkeeping split of the survival deficit into a pointer-diagonal term and an off-sector remainder. In addition, Proposition 6.14 and Corollary 6.16 now show that the foreground-background part of the model-independent programme closes conditionally once four tail inputs are granted: approximate tail optimality, exchange-cover plus persistence, tail-passive background survival, and an exchange-efficiency gap, together with the block-locality defects for the KL and control terms. Corollary 6.17 then reduces the remaining diagonal product-defect problem to finitely many ordered within-block prefix couplings. The remaining open issue is therefore to derive those tail hypotheses together with the prefix-level coupling and locality bounds model-independently from the broken-phase output and the full T-DOME survival identity rather than assume them. Beyond the restricted regime, the unresolved analytic obstacles are explicit: cross-channel correlations need not decouple, pointer spectra need not remain finite or discrete, and the Hilbert–Schmidt control used here is not available in general Type III infinite-dimensional settings, where a predual reformulation may be necessary. The pointer-sector existence input PS1 is treated here only as an exact-regime idealization; what remains genuinely open is the model-independent alignment step PS2 and the induced ordered-selection step PS3.

3. **Comparison with typed HAFF objects.** The present paper proves inclusion only into the exact fixed sector under FSA and MOD. Equality with that sector remains open, and no canonical equivalence with probe-accessible effects or the environmental record algebra is asserted.

4. **Bridge to Paper I admissibility.** To feed directly into Paper I, one must still verify that the algebra produced by the lower bridge satisfies the additional admissibility hypotheses used there, beyond the already-closed faithfulness condition. More precisely, the remaining interface is to supply the independent dynamical modular-decoherence scaling axiom (C) together with an operational shell-generation/closure interface $\{\mathcal{A}_*(\tau)\}_{\tau>0}$ around the stable-hull output $\mathcal{A}_* := \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$, and one-scale cyclicity of the corresponding shell-annulus algebra
$$\mathcal{D}_\tau := \mathcal{A}_*(\tau) \cap \mathcal{A}_*'.$$
In Paper I language these are exactly the remaining inputs $(C)+(Amb)+(CN_\exists)$. Foundation I does not derive the shell-generation rule or its covariance. The latter is to be derived in Paper I from the independently stated dynamical axiom (C), once membership and closure are supplied. The shell-family side has only partially internalized: Proposition 7.6 closes anchored relative-shell existence around $\mathcal{A}_*$. What remains open is the canonical and calibrated nontrivial shell family. The naive scalar-threshold route now has an exact blocker: fixed thresholds are compatible with modular transport but not with exact core calibration, vanishing thresholds give exact core calibration but not automatic modular transport, and the only scalar profile with both properties is the trivial zero-threshold profile, which collapses the shells back to $\mathcal{A}_*$. More sharply, even the fixed-threshold branch now reduces to finding a scale-invariant nontriviality window between core collapse and ambient collapse, together with uniqueness of maximal shells inside that window; exact calibration is a separate issue beyond that.

These open problems are real, but they are no longer vague. Foundation I is now separated into one conditional source-selection problem, one theorem-level thermodynamic packaging result, one theorem-level stable-hull transfer construction, and one remaining foreground-realization/comparison problem.

9 Conclusion

The purpose of this paper has been calibration, not inflation. Foundation I is neither absent nor complete. It is partially available in theorem-level form, partially available as an explicit construction problem, and partially still conditional.

The main positive result is that the T-DOME I–II chain can already be written as a formal Foundation I proposition, and that the first restricted PS2 bottleneck can now be closed at theorem level in a first restricted regime. In the product-classical, decoherence-dominated pointer regime, pointer coarse-graining yields fragment-local omission losses, independent fragment records force orthogonality, semigroup decay plus explicit Hilbert–Schmidt continuity constants bound the coherence remainder, and a sufficiently large weight gap forces exact top-k^* pointer selection. On the algebraic side, the first operator-algebraic realization of the foreground sector can now be extended to a well-defined minimal modular-decoherence stable hull, yielding a faithful von Neumann subalgebra together with the inclusion

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\mathrm{HAFF}}.$$

At the level of future PS2 work, the paper now also fixes the bookkeeping split of the survival deficit into a pointer-diagonal term and an off-sector remainder, and it closes the first foreground-background decoupling step conditionally: if the tail distortion localizes, the exchange-weighted defect persists, the background diagonal survival becomes tail-passive, the exchange efficiency ratio stays below the squared-loss threshold, and the remaining KL/control locality defects are small, then the broken phase forces tail-small diagonal foreground-background coupling and hence tail-small foreground-background mutual information. Corollary 6.17 then shows that the unresolved product-defect part is no longer a monolithic statement: it is a finite family of ordered within-block prefix decoupling problems. The remaining model-independent problem is therefore no longer vague: it is to derive those tail hypotheses and the ordered prefix-level coupling/locality bounds, and thereby to control the full diagonal product defect as a theorem-level output of the broken phase. On the Paper I side, the admissibility interface is now sharpened in the same way: faithfulness is already closed, ambient scaling covariance would transfer the dynamical axiom (C), and the remaining regularity burden is exactly the existence of a Foundation I shell family and one cyclic shell-annulus scale. Even the shell-family side is no longer a black box: anchored relative shells around $\mathcal{A}_*$ exist, but their upgrade to a canonical nontrivial shell family currently stops at the scalar-threshold trilemma between transport, exact core calibration, and nontriviality. Within the fixed-threshold route, this has already reduced further to a scale-invariant nontriviality window and uniqueness problem, plus a separate calibration theorem. The main negative result is equally important: nothing in the present note licenses the stronger claim that the source datum has already been shown to generate the full accessible algebra uniquely, nor that PS2 has been derived model-independently from the broken-phase output. Those remain the unfinished parts of the unified-source programme.

In that sense, the current status is simple. The thread can be strung, but only honestly: conditional at the source end, inclusion-first at the algebra end, and theorem-backed only where the intermediate steps have actually been closed.

A External Genesis Inputs (Self-Contained Statements)

The following list records the external Genesis-side inputs used in the paper in a self-contained form. The point is not to reproduce their full proofs, but to state exactly what is imported from HAFF and T-DOME.

1. **HAFF ambient datum.** The ambient setting is a Type III_1 von Neumann algebra in standard form $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, with Ω cyclic and separating for $\mathcal{A}_U$, together with a pointwise σ-weakly continuous semigroup of normal completely positive unital maps $\{\mathcal{E}_t\}_{t\geq 0}$ on $\mathcal{A}_U$. This is the source datum formalized in Section 2.

2. **T-DOME I projected record dynamics.** Given a persistent Nakajima–Zwanzig projection $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$, the projected predual dynamics is analyzed in a non-Markovian kernel form

$$\partial_t \omega_{\mathcal{P},t} = \int_0^t \mathcal{K}_*(t,s)\, \omega_{\mathcal{P},s}\, ds \;+\; \text{inhomogeneous term},$$

   or equivalently in the dual memory-kernel formulation used in Section 2. Foundation I uses only the existence of such a kernel and a persistent record process, not an explicit closed-form solution.

3. **T-DOME I survival identity and distortion functional.** The survival bookkeeping imported into the PS2 analysis is the identity

$$\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}} - \beta\, \Delta\langle H_{\mathrm{ctrl}}\rangle,$$

   with frame comparisons measured in the squared-loss distortion

$$D(\mathcal{F}_I) := \mathbb{E}_\xi\Big[\big(\mathcal{S}_{\mathrm{full}}(\xi) - \mathcal{S}_{\mathcal{F}_I}(\xi)\big)^2\Big].$$

   The diagonal/off-sector split and all restricted PS2 estimates are built from this identity rather than assumed independently.

4. **Finite-coordinate compression model motivated by T-DOME II.** T-DOME II motivates a coordinate-coding problem, but its universal computational-ceiling and SSB-necessity claims are not imported. Proposition 4.1 instead assumes a rate charge, strict improvement up to the feasible rank, and an irreducible group action. Only in that supplied FD/I model does $k^* = \lfloor \dot{\mathcal{C}}_{\mathrm{budget}}/h_\mu \rfloor$ and the selection of a proper foreground subspace follow.

5. **Decoherence-side pointer sector (exact-regime idealization).** In the exact dephasing-dominated regime, robust classical records are carried by a commuting pointer sector. Foundation I uses this as the PS1 input: a commuting family of exact fixed-point projections for $\{\mathcal{E}_t\}$ generating a classical pointer algebra inside $\mathcal{A}_U$. This is an idealization of the approximate einselection and redundant-record structure described in the quantum Darwinism literature [9, 10], not a consequence of generic decoherence alone.

6. **Exact fixed sector, not a universal accessibility object.** The inclusion comparison uses $\mathcal{A}_{\mathrm{fix}} = \mathrm{Fix}(\mathcal{E})$ only under FSA and MOD. This object is an exact noiseless/fixed sector. It is not identified with the probe-accessible operator system or the environmental pointer/record algebra.

7. **T-DOME III / Paper III calibration distinction.** Foundation I uses T-DOME III and Paper III only as contextual support for what is *not* claimed: the foreground complexity k^* is not identified with Paper III's statistical complexity parameter n, and the present paper does not attempt a direct derivation of Paper III's self-description bounds from the source-side chain.

B Conditional Input Ledger

For ease of reference, the remaining conditional interfaces are listed here in one place.

1. **Source selection.** Lemma 3.2 is conditional on the existence of a persistent Nakajima–Zwanzig projection producing the projected record kernel. This is the observer-emergence bottleneck. No general semigroup-gap, CP-divisibility, or kernel-decay criterion is asserted here: each would require a specified persistence functional and can fail to capture the non-Markovian record regime. A rigorous sufficient criterion remains open rather than being installed for downstream use without proof.

2. **Pointer alignment package.** The foreground realization problem separates into:

   (PS1) existence of a decoherence-selected exact fixed-point pointer sector (exact dephasing-dominated regime idealization, not a consequence of generic decoherence);

   (PS2) identification of the T-DOME sufficient-statistic coordinates with the pointer-sector coordinates;

   (PS3) ordered pointer-channel selection by survival relevance.

   PS1 is an exact-regime idealization of the decoherence-side input; the genuine model-independent open problem is PS2, with PS3 downstream from it.

3. **Exact fixed-sector structural inputs.** The main text keeps the following conditions separate because none is a consequence of the bare Type III source datum:

   (FSA) $\mathrm{Fix}(\mathcal{E})$ is closed under the ambient product and is a von Neumann subalgebra;

   (MOD) that fixed algebra is invariant under the ambient modular flow;

   (EXP) when an expected subalgebra is required, an ω-preserving faithful normal conditional expectation exists;

   (PTR) the separately specified environmental pointer algebra is contained in the exact fixed algebra in the idealized noiseless regime.

   Faithfulness of ω alone does not establish MOD. In the faithful standard-form regime, Takesaki's theorem relates MOD to an ω-preserving expectation only after FSA supplies the subalgebra to which the theorem applies [12, 13].

4. **Restricted PS2 regime assumptions.** The theorem-level restricted PS2 closure is proved only in the product-classical, finite-dimensional faithful pointer regime stated in Section 6. In particular, the restricted closure does not yet treat general cross-channel correlations, continuous pointer spectra, or the genuinely infinite-dimensional Type III estimate problem.

5. **Conditional foreground-background decoupling inputs.** Proposition 6.14 requires the following named inputs on the tail regime: approximate tail optimality, exchange-cover plus persistence, tail-passive background survival, and an exchange-efficiency gap, together with the KL/control locality defects used in Corollary 6.16.

6. **Abstract compatibility of the four tail packages.** An engineered finite classical Markov witness realizes the four tail packages simultaneously. This is an FD/I audit witness establishing joint mathematical compatibility, not a physical derivation and not a Type III shell model.

7. **Natural physical nonvacuity remains open.** No natural source/control model is known to derive all four packages jointly, and no known model simultaneously realizes those packages and the modular shell interface. Proposition 6.14 therefore remains conditional despite the compatibility witness.

8. **Conditional full product-defect inputs.** Corollary 6.17 adds the ordered within-block prefix-chain analogues of the same decoupling/locality inputs. The full diagonal product-defect problem is therefore reduced to finitely many named prefix couplings rather than left as one undifferentiated gap.

9. **Paper I admissibility inputs.** The lower output feeds into Paper I only conditionally on three further interfaces: the independent dynamical covariance axiom (C), an operational shell membership/closure rule around the stable hull, and one-scale shell-annulus cyclicity. Proposition 7.6 closes only an anchored existence problem. It does not derive the Paper I shell interface. In particular, covariance is not placed into that interface; Paper I must derive shell scaling from (C).

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